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using System;
using System.Collections.Generic;

namespace SamplePrograms
{
    public class RomanNumeral
    {
        private static readonly Dictionary<char, int> RomanDecMapping =
new Dictionary<char, int>()
        {
            ['M'] = 1000,
            ['D'] = 500,
            ['C'] = 100,
            ['L'] = 50,
            ['X'] = 10,
            ['V'] = 5,
            ['I'] = 1,
        };

        private static int RomanToDecimal(string roman, int total=0)
        {
            if (roman.Length < 1)
                return total;
            var romanArray = roman.ToCharArray();
            if (romanArray.Length == 1)
                return total + RomanDecMapping[romanArray[0]];

            var romanVal = RomanDecMapping[romanArray[0]];
            var nextRomanVal = RomanDecMapping[romanArray[1]];
            if (romanVal < nextRomanVal)
                return RomanToDecimal(roman.Substring(1), total -
romanVal);

            return RomanToDecimal(roman.Substring(1), total + romanVal);
        }

        public static void Main(string[] args)
        {
            if (args.Length < 1)
            {
                Console.WriteLine("Usage: please provide a string of
roman numerals");
                Environment.Exit(1);
            }
            try
            {
                Console.WriteLine(RomanToDecimal(args[0].ToUpper()));
            }
            catch (KeyNotFoundException)
            {
                Console.WriteLine("Error: invalid string of roman
numerals");
                Environment.Exit(1);
            }
        }
    }
}

```

}
}
}